



Customizable AI-Enabled ADAS Demonstrator

Capstone Project | Group 66

Department of Systems and Computer Engineering
Course: SYSC 4907
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The 1.4 Billion Vehicle Safety Gap

The Problem

1.4 Billion

Total vehicles globally without modern safety features.



3,732 Traffic deaths in Canada (2023)

- **The Reality:** Modern cars have “eyes.” Older cars do not.
- **Human Factor:** Failure to perceive hazards causes the majority of accidents.

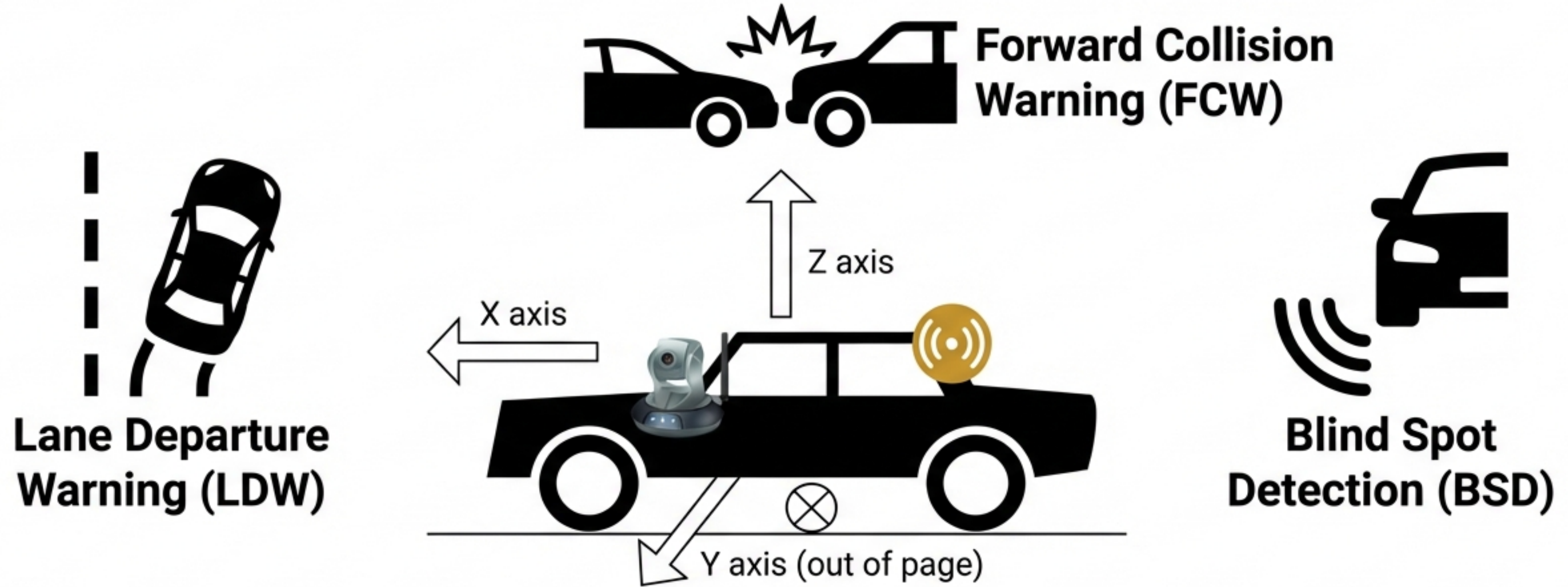
The Context



Demographic Constraint: Older vehicles are often owned by lower-income drivers. Expensive retrofits are inaccessible.

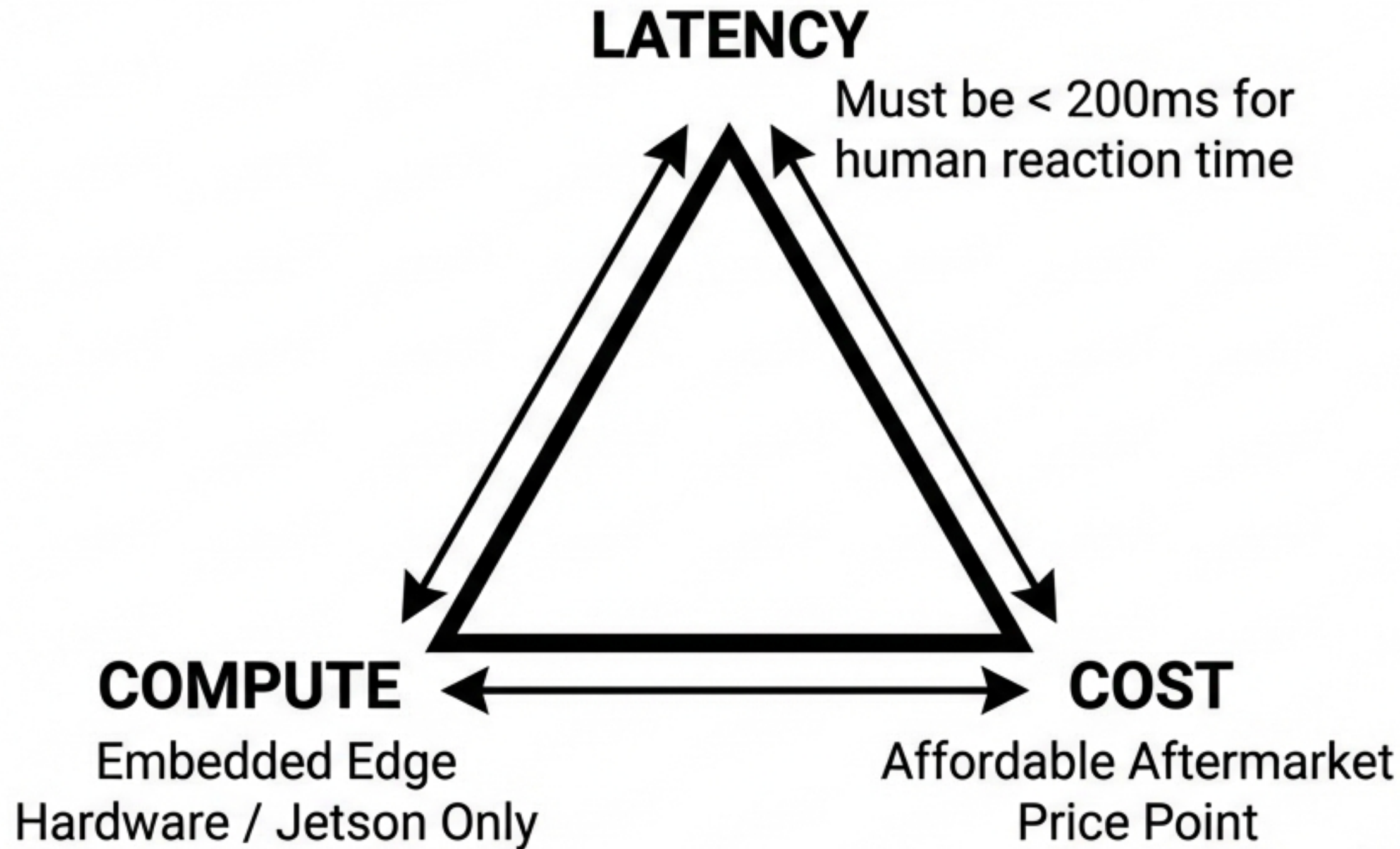
Current State: Existing aftermarket solutions are often unconnected, “dumb” devices lacking true sensor fusion.

Democratizing Safety Technology



Mission: A low-cost, aftermarket retrofit for the 2012 Honda Civic Coupe Si. System provides SAE Level 0/1 Warnings (No Brake Control).

The Triangle of Engineering Constraints



NFR29: Max latency from decision to app render $\leq 100\text{ms}$ (p95).

NFR30: Bluetooth link must maintain 20 Hz alert throughput.

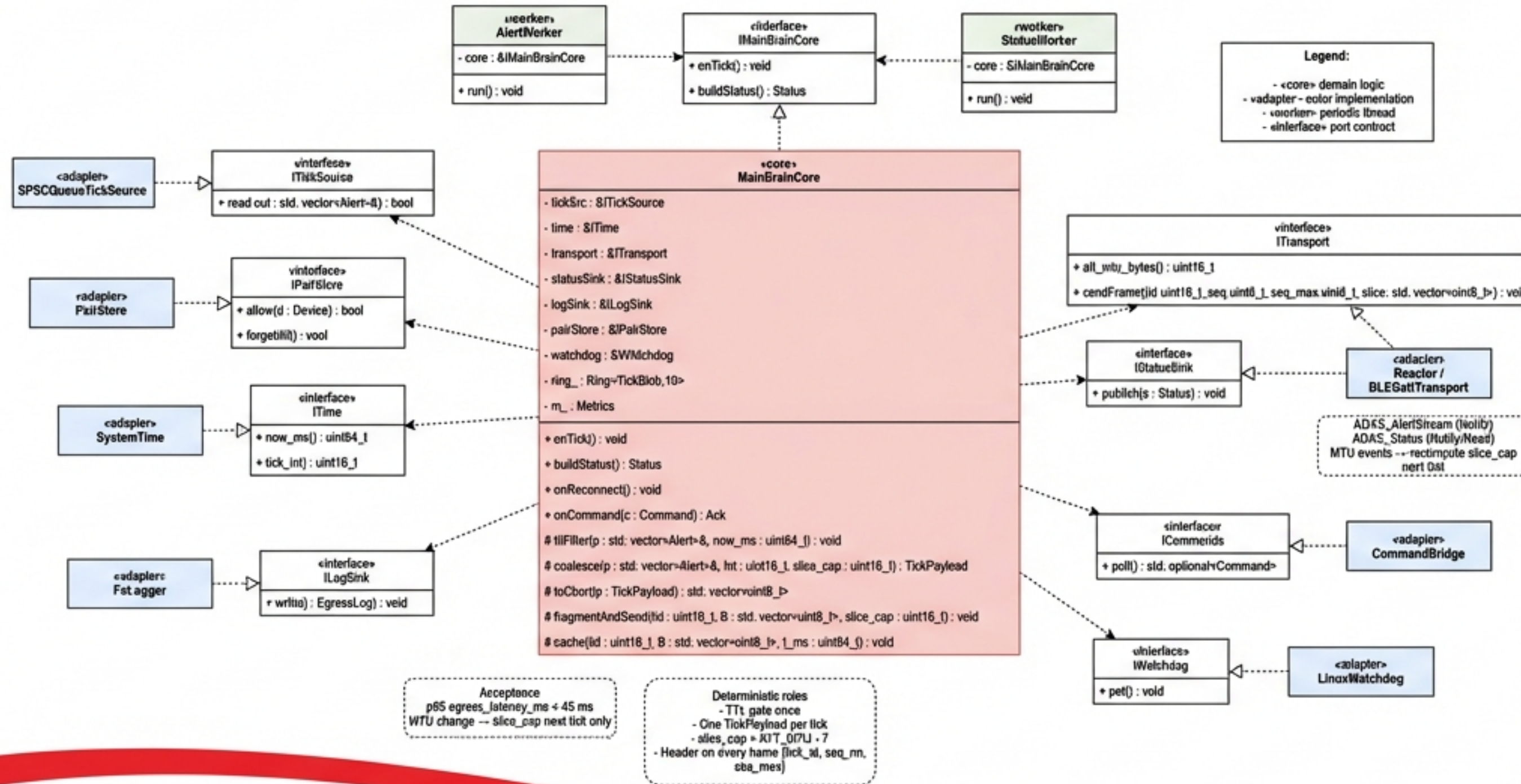
Hardware Implementation & The “Pivot”

- **Compute:** Okdo Jetson Nano C100 (Replaced Orin Nano due to supply chain)
- **Vision:** 4x InnoMaker U20CAM-1080P (360° View)
- **Radar:** 1x OmniPreSense OPS243 (Front), 2x DFRobot C4001 (Blind Spots)
- **IMU:** 1x Adafruit BNO085 (9-DOF)

Engineering Challenge: Mapping 3 independent UART interfaces on non-standard carrier carrier board.

3.3V	1		2	5.0V
I2C1_SDA	3		4	5.0V
I2C1_SCL	5		6	GND
GPIO09	7		8	UART1_TXD
GND	9		10	UART1_RXD
UART1_RTS*	11		12	I250_SCLK
SPI1_SCK	13		14	GND
GPIO12	15		16	SPI1_CS1*
3.3V	17		18	SPI1_CS0*
SPI0_MOSI	19		20	GND
SPI0_MISO	21		22	SPI1_MISO
SPI0_SCK	23		24	SPI0_CS0*
GND	25		26	SPI0_CS1*
I2C0_SDA	27		28	I2C0_SCL
GPIO01	29		30	GND
GPIO11	31		32	GPIO07
GPIO13	33		34	GND
I250_FS	35		36	UART1_CTS*
SPI1_MOSI	37		38	I250_DIN
GND	39		40	I250_DOUT

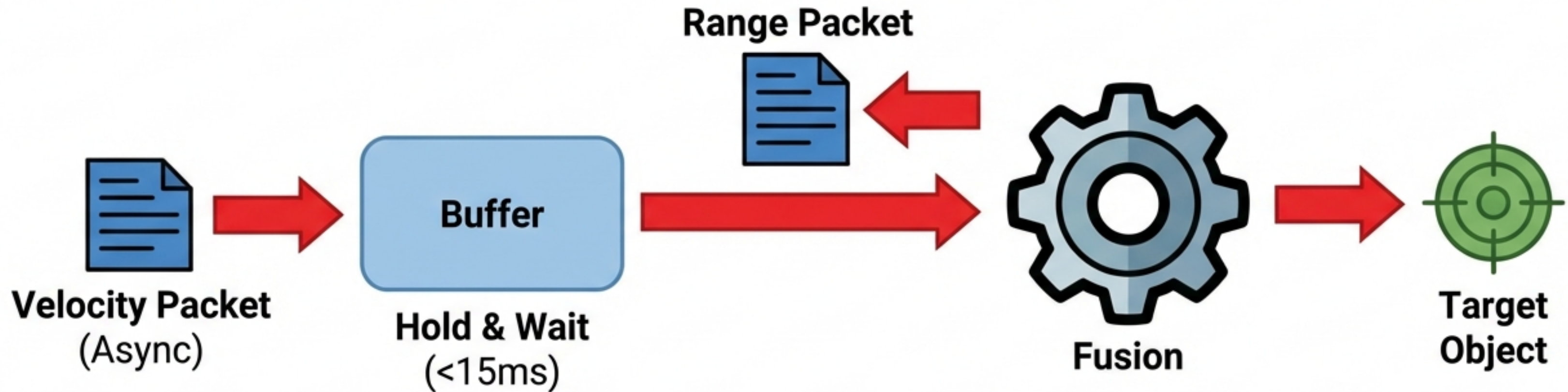
Hexagonal Architecture (Ports & Adapters)



Why? Decouples Domain Logic from I/O. Allows sensor hardware swaps without rewriting collision algorithms (FR16).



Solving the 'Split Packet' Problem

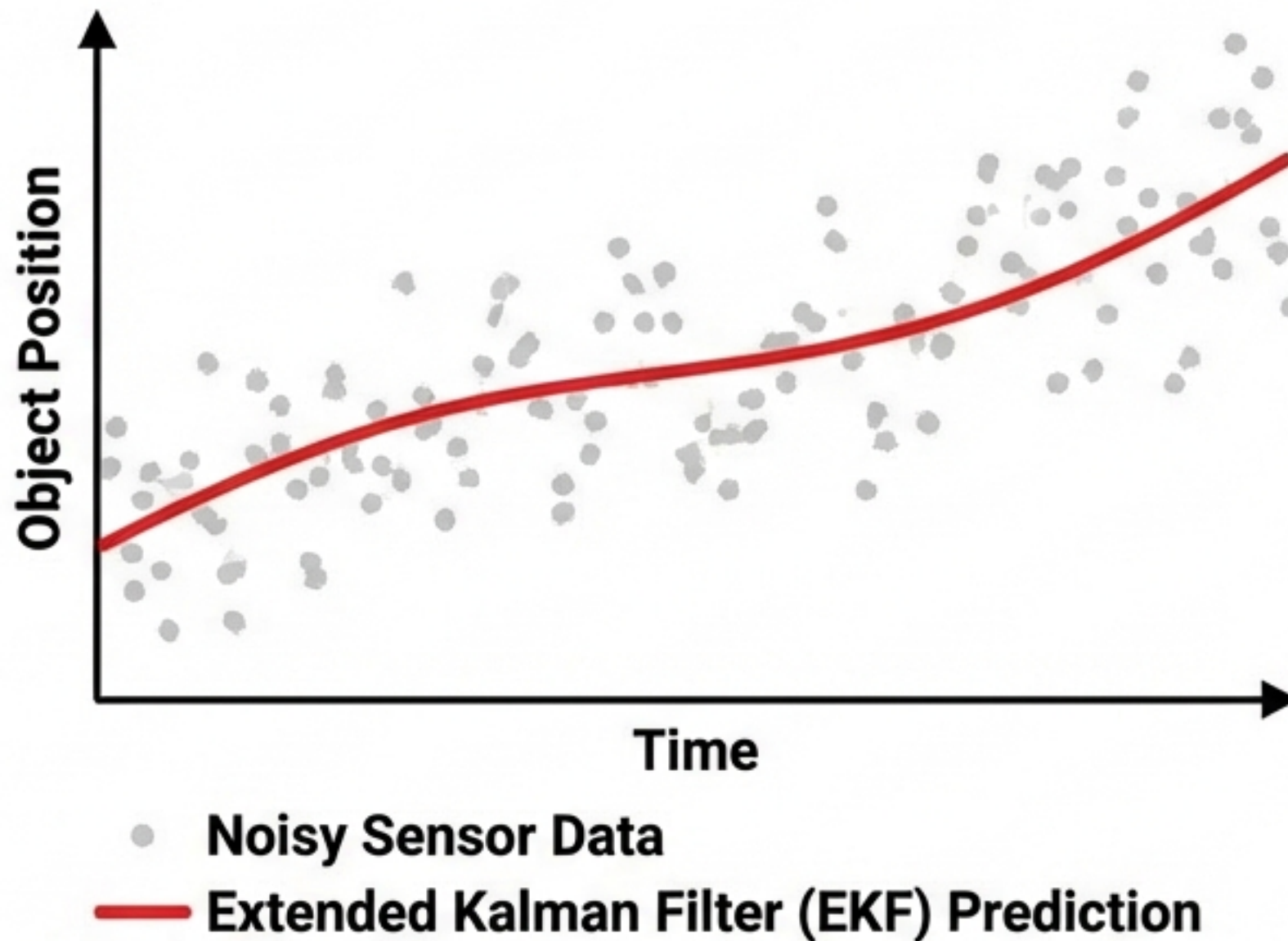


Challenge: The OPS243 Radar lacks Azimuth and sends Range/Velocity asynchronously.

Solution: Buffer velocity, wait for range, fuse into single target.

Result: Radar data (Range/Speed) + Camera data (Object Class) = 2D Threat Location.

The Mathematics of Safety (Fusion & Tracking)



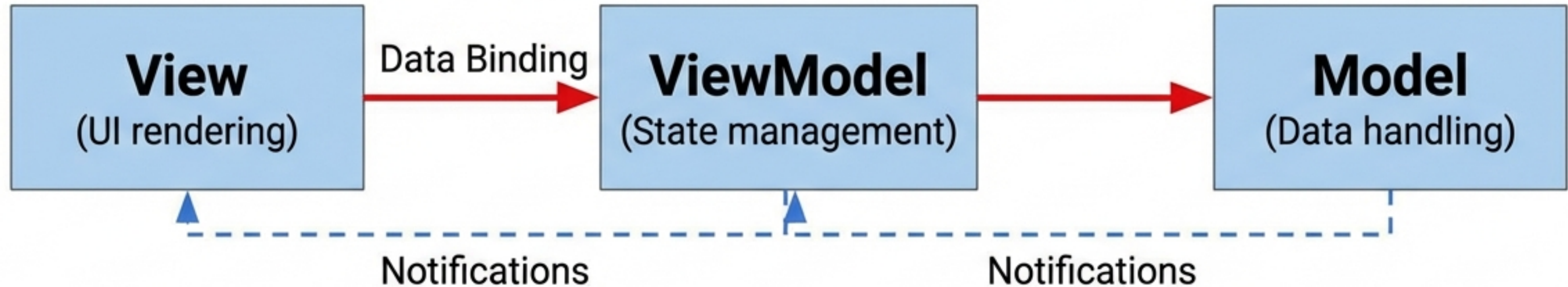
Extended Kalman Filter (EKF):

Estimates true state $\{x, y, vx, vy\}$ from noisy inputs (FR50).

- **Gating:** Uses Mahalanobis Distance to reject false positives (FR35).
- **Decision Logic:** Warning triggers only if Time-To-Collision (TTC) $< 2.5s$ AND Relative Velocity is Negative (Closing).

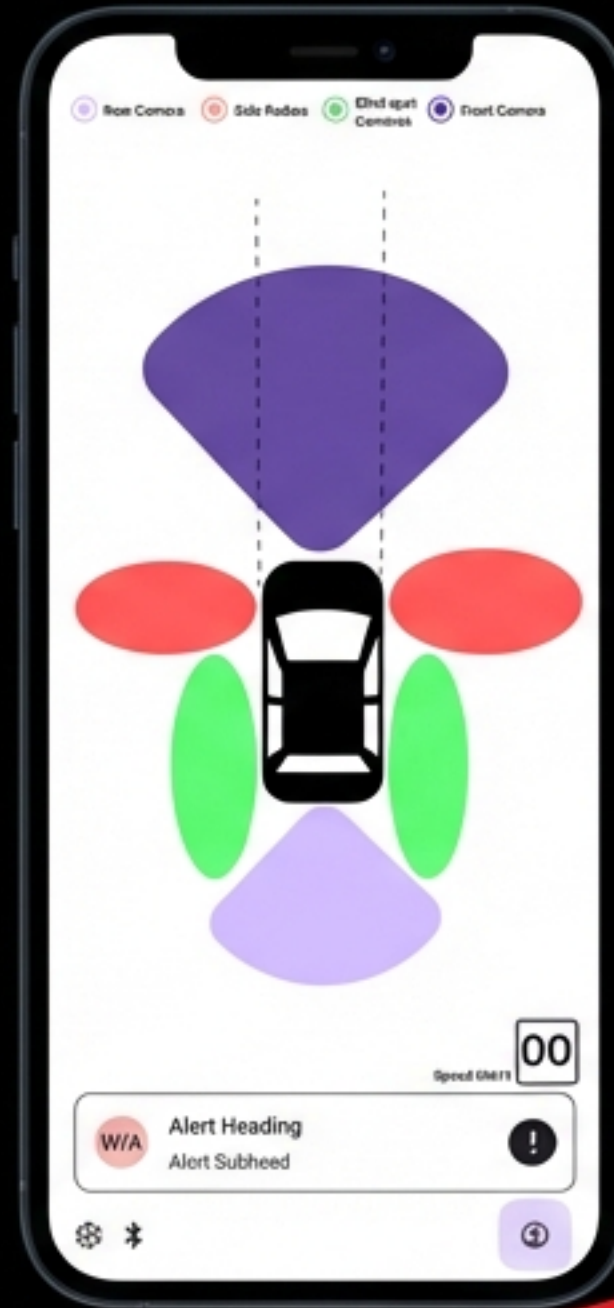


Mobile Architecture: MVVM Pattern



- **Pattern:** Model-View-ViewModel decouples high-frequency data from UI.
- **Benefit:** Prevents 'App Not Responding' (ANR) by processing 20Hz streams in background threads.
- **Tech Stack:** Kotlin, Gradle, CBOR (Binary Object Representation for speed).

The 'Dark Cockpit' UX Philosophy



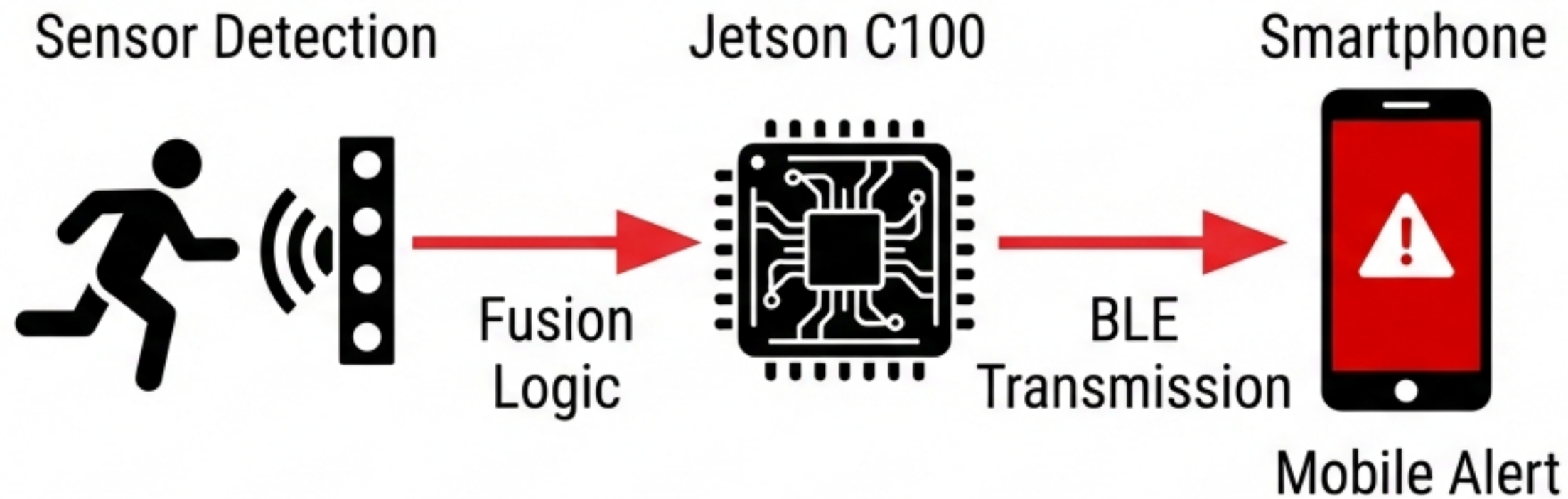
Philosophy: Minimize cognitive load. High contrast, low glare.

Alert Hierarchy:

- Green: Monitoring / Safe
- Red: Critical ($TTC < \text{Threshold}$)

Safety Lock: Interactive inputs disabled when Speed > 0 (FR73).

Verification: The Vertical Slice



Sensor Detection -> Fusion Logic -> BLE Transmission -> Mobile Alert

SUCCESS

**Critical Path Latency:
< 200ms**

Confirmed safety margin
for human reaction time.

Project Timeline & Execution

	Task Name	Oct	Nov	Dec	Jan	Feb	March
1	Phase 1: Architecture & Procurement						
2	Phase 2: Full Implementation (Current)						
3	Phase 3: Field Testing & Poster Fair						

Current Status: 9 Parallel Workstreams (AI, BLE, DevOps) active.
Verification: Continuous Integration (CI) with fault injection testing.

Cross-Functional Engineering Team

Software Engineering	Electrical Engineering	Computer Systems
Ajen Srisivapalan	Rami Mrad	Ryan Page
Damon Ricci	(Sensor Hardware)	(Embedded/DevOps)
Jason Keah (App/UI)		
John Guo (Radar/Physics)		

Supervisor: Dr. Mohamed Atia

Path to Completion

- ✓ Architecture & Vertical Slice
- Full Vehicle Integration (Mounting on Civic)
- Field Calibration of Extrinsic
- Validation (Recall ≥ 0.80)

Target: A fully road-ready demonstrator for March Final Review.

Closing the Safety Gap

Intelligent Safety for the 1.4 Billion
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